



User Manual

by Brandon Falk

Introduction

TE Tool is designed to batch-check media files for a series of technical specifications used for Broadcast Masters or SVOD. It runs without altering the original files.

All results are based on automated analysis using FFmpeg and related utilities. Some checks may require manual verification, but this will hopefully help speed things up.

Running A Check

1. Open the TE tools App
2. Drag the files you want checked onto the open application (you can check more than one at a time)
3. Select what kind of test to run
 - The drop-down menu will appear allowing you to select SVOD or Broadcast Master. Make your selection.
 - Adjust the check-boxes based on the requirements, the click "OK"
4. Wait for results
5. Review results - Once complete, there will be a notification sound, and the results will show for each file.

User Manual

TE TOOL – CHECK MAP AND DESCRIPTIONS



TEST TYPE SELECTION

- SVOD: check groups available are Phase 1, I Frame, Phase 2, Luma IRE, True Peaks, Advanced Luma IRE, Advanced Chroma.

- Broadcast Master: check groups available are Phase 1, I Frame, Caption Profanity, Phase 2, Luma IRE, True Peaks, Advanced Luma IRE, Advanced Chroma.

PHASE 1 (SVOD and Broadcast Master)

What runs

- File Type: Reads the extension. Pass when .mov.

Limitations: Only checks the extension, not container compliance.

- Video Bit Rate: Reads bit_rate from the video stream and prints Mb per second.

Limitations: Some files do not carry accurate bit_rate metadata.

- Video Codec: Reads codec_tag_string. Pass when tag is apch which maps to ProRes 422 HQ.

Limitations: Other ProRes profiles will fail this specific rule even if they are valid for a different spec.

- Resolution: Reads width and height. Pass when 1920 by 1080.

Limitations: Non-square pixel sources are not fully evaluated.

- Aspect Ratio: Uses display_aspect_ratio when present, otherwise estimates from width and height.

Limitations: DAR missing or unusual PAR may require manual scopes.

- Frame Rate: Converts r_frame_rate to a float. SVOD intent is 23.98. Broadcast Master intent is 29.97.

Limitations: Variable frame rate may round oddly. Values are printed to two decimals.

- Chroma Sampling: Pass when pixel format starts with yuv422 which indicates 4:2:2.

Limitations: This does not validate chroma siting or conversion paths.

- Scan Type: Pass when field_order is progressive.

Limitations: Some encoders do not tag interlace fields correctly.

- Audio Codec: Pass when codec name starts with pcm.

- Audio Bit Rate: Expects 1152 Kb per second for mono and 2304 Kb per second for stereo based on 48 kHz 24 bit PCM.

Limitations: Derived from metadata. If the muxer omits it, the rule may fail.

- Audio Sample Rate: Pass when 48000.

- Audio Bit Depth: Tries tag first then infers from sample_fmt. Pass when 24.

Limitations: Some files hide bit depth. The script makes a best guess.

- Phase 1 (Broadcast Master Only) : Timecode Format

Checks if the file is drop frame. Pass when drop frame is detected.

Limitations: Relies on timecode tags. Missing tags will fail this rule.

User Manual

TE TOOL – CHECK MAP AND DESCRIPTIONS



IFRAME

- I Frame Checker: Uses packet flags to see if every packet is marked I. Then inspects the first 5000 frames for P or B types. Pass when no non-I frames are seen. Limitations: Only the first 5000 frames are slice-verified. Some codecs expose flags differently.

PHASE 2 – SVOD

- Head Dark Frame: Samples a frame near the start using signalstats YAVG against a codec family threshold. Pass when under threshold. Limitations: This is a brightness check only.

- Tail Dark Frame: Samples a frame near the end using the same method. Pass when under threshold. Limitations: This is a brightness check only.

- Stereo Only: Pass when there is exactly one audio stream and it reports two channels. If channels are missing, layout text is used to infer stereo. Limitations: Multichannel or mono files fail even if a stereo fold-down exists elsewhere.

- Dynamic Stereo: Spot checks four short windows. Computes L minus R RMS and reports the largest imbalance plus a simple left or right label. Pass when L minus R RMS is greater than minus 50 dB. Limitations: This is a coarse spot check. It does not replace a full phase or correlation review.

- Loudness: Decodes audio to 24 bit WAV and runs loudnorm in analysis mode. Reports Integrated LUFS. Pass when between minus 25 and minus 22. Limitations: Very short files may not produce reliable LUFS.

- Timecode Start: Pass when timecode starts at 00:00:00:00 or 00:00:00;00. Limitations: Missing or malformed tags cause a Not detected note.

PHASE 2 – BROADCAST MASTER

- Stereo Only: Same rule as SVOD.
- Loudness: Same rule as SVOD.
- Timecode Start: Pass when timecode starts at 00:58:30;00 or 00:58:30:00.
- Segment Markers: Uses ExifTool to read timeline markers if present and prints a small table. Limitations: Only reads markers that were embedded. None means no table.

- Total Run Time: Uses frame count from ExifTool when available or ffprobe otherwise. Converts to drop frame timecode with a 30 fps base. Limitations: Assumes 29.97 style counting. Non-29.97 content may show a misleading DF value.

- 608 Captions Present: Pass when an eia_608 subtitle stream is present. Limitations: Presence only. No content validation.

- 708 Captions Present: Pass when an eia_708 subtitle stream is present. Limitations: Presence only. No content validation.

User Manual

TE TOOL – CHECK MAP AND DESCRIPTIONS



- Caption Profanity (Broadcast Master group list includes this): Extracts the first subtitle stream to SRT and runs cussbuster2000.py. If the Python script outputs a profanity report, TE Tool prints it and marks a fail.

Limitations: Keyword based. Only the first subtitle stream is scanned. False positives and misses are possible.

- Luma IRE: Samples every 24th frame using YMIN and YMAX. Converts to IRE based on full or limited range for the pixel format. Pass when min and max stay within minus 1 to 101 IRE. Reports min and max with timestamps.

Limitations: Sampling cadence can miss very short spikes.

- True Peaks: Splits stereo into left and right WAV, runs loudnorm summary on each leg, reads Input True Peak. Pass at or under minus 2.5 dBTP. Close warning between minus 2.5 and minus 1.5 dBTP.

Limitations: Based on ffmpeg loudnorm peak model. Treat close calls as advisory.

- Advanced Luma IRE: Same math as Luma IRE but samples every frame. Uses a one hour header timecode offset when it can detect it.

Limitations: Slower on long files.

- Advanced Chroma: Runs signalstats in the native pixel format, collects global min and max for Cb and Cr, shows approximate 8 bit values, and flags excursions against TV legal when the video is limited range.

Limitations: Does not skip the header.

If you have any questions or suggestions, please reach out to me,
Brandon Falk